Suggested problems §20.1: 3-9 odd

Suggested problems §20.2: 1-11 odd

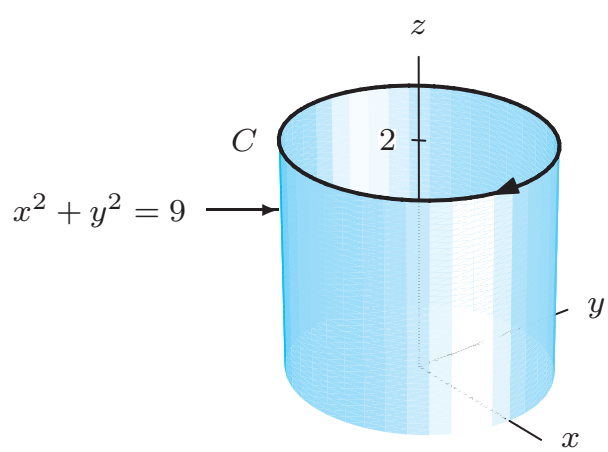
Problem 1: Compute the curl $\nabla \times \vec{F}$ for each of the following vector fields.

(a) $\vec{F}(x, y, z) = 3x\vec{i} - 5z\vec{j} + y\vec{k}$

(b) $\vec{F}(x, y, z) = \langle e^x, z\cos y, e^{y^2} \rangle$

Problem 2: The figure to the right shows an open cylindrical can S standing on the xy-plane. (S has a bottom and sides, but no top.) If S is oriented outward and downward, compute

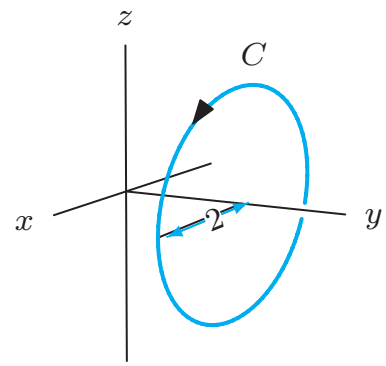
$$\int_S \nabla \times \langle -y, x, z \rangle \cdot d\vec{A}.$$



Problem 3: Compute

$$\oint_C (-z\vec{i} + y\vec{j} + x\vec{k}) \cdot d\vec{r}$$

where C is the circle of radius 2 shown to the right.



Problem 4: Compute

$$\oint_C \langle x^2, y^2, z^2 \rangle \cdot d\vec{r}$$

where C is the unit circle in the xz-plane, oriented counterclockwise when viewed from the positive y-axis.

Problem 5: Compute

$$\oint_C (-y\vec{i} + x\vec{j} + z\vec{k}) \cdot d\vec{r}$$

where C is the circle of radius 2 in the plane $x + y + z = 3$, centered at $(1, 1, 1)$ and oriented clockwise when viewed from the origin.